

README

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1 toddwint/ntp

1.1 Info

ntp docker image for simple lab testing applications.

Docker Hub: <https://hub.docker.com/r/toddwint/ntp>

GitHub: <https://github.com/toddwint/ntp>

1.2 Overview

Docker image for hosting an NTP server fudging a stratum 0.

Pull the docker image from Docker Hub or, optionally, build the docker image from the source files in the build directory.

Create and run the container using `docker run` commands, `docker compose` commands, or by downloading and using the files here on github in the directories `run` or `compose`.

Manage the container using a web browser. Navigate to the IP address of the container and one of the HTTPPORTs.

NOTE: Network interface must be UP i.e. a cable plugged in.

Example `docker run` and `docker compose` commands as well as sample commands to create the macvlan are below.

1.3 Features

- Ubuntu base image
- Plus:
 - `fzf`

- iproute2
- iputils-ping
- ntp
- python3-minimal
- rsyslog
- tmux
- tzdata
- [ttyd](#)
 - View the terminal in your browser
- [frontail](#)
 - View logs in your browser
 - Mark/Highlight logs
 - Pause logs
 - Filter logs
- [tailon](#)
 - View multiple logs and files in your browser
 - User selectable tail, grep, sed, and awk commands
 - Filter logs and files
 - Download logs to your computer

1.4 Sample commands to create the macvlan

Create the docker macvlan interface.

```
docker network create -d macvlan --subnet=192.168.10.0/24 --gateway=192.168.10.254 \
--aux-address="mgmt_ip=192.168.10.2" -o parent="eth0" \
--attachable "ntp01"
```

Create a management macvlan interface.

```
sudo ip link add "ntp01" link "eth0" type macvlan mode bridge
sudo ip link set "ntp01" up
```

Assign an IP on the management macvlan interface plus add routes to the docker container.

```
sudo ip addr add "192.168.10.2/32" dev "ntp01"
sudo ip route add "192.168.10.0/24" dev "ntp01"
```

1.5 Sample docker run command

```
docker run -dit \
--name "ntp01" \
--network "ntp01" \
--ip "192.168.10.1" \
-h "ntp01" \
-p "192.168.10.1:123:123/udp" \
-p "192.168.10.1:8080:8080" \
-p "192.168.10.1:8081:8081" \
```

```

-p "192.168.10.1:8082:8082" \
-p "192.168.10.1:8083:8083" \
-e TZ="UTC" \
-e MGMTIP="192.168.10.2" \
-e GATEWAY="192.168.10.254" \
-e HTTPPORT1="8080" \
-e HTTPPORT2="8081" \
-e HTTPPORT3="8082" \
-e HTTPPORT4="8083" \
-e HOSTNAME="ntp01" \
-e APPNAME="ntp" \
--cap-add=NET_ADMIN \
"toddwint/ntp"

```

1.6 Sample docker compose (compose.yaml) file

```

name: ntp01

services:
  ntp:
    image: toddwint/ntp
    hostname: ntp01
    ports:
      - "192.168.10.1:123:123/udp"
      - "192.168.10.1:8080:8080"
      - "192.168.10.1:8081:8081"
      - "192.168.10.1:8082:8082"
      - "192.168.10.1:8083:8083"
    networks:
      default:
        ipv4_address: 192.168.10.1
    environment:
      - MGMTIP=192.168.10.2
      - GATEWAY=192.168.10.254
      - HOSTNAME=ntp01
      - TZ=UTC
      - HTTPPORT1=8080
      - HTTPPORT2=8081
      - HTTPPORT3=8082
      - HTTPPORT4=8083
      - APPNAME=ntp
    privileged: true
    cap_add:
      - NET_ADMIN
    tty: true

```

```
networks:  
  default:  
    name: "ntp01"  
    external: true
```